

Aruv Dand

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EDUCATION

University of Illinois Urbana-Champaign

Expected Grad: May 2027

Bachelor of Science + Master of Computer Science, Certificate in Data Science

GPA: 3.97/4.0

RELEVANT COURSEWORK

Courses: Data Structures, Algorithms, Computer Architecture, Numerical Methods, Artificial Intelligence, Systems Programming, Stats & Probability, Calculus 3, Linear Algebra, Applied ML, Networks, Compilers, Audio Computing

Awards: Engineering Dean's List, National Merit Finalist Scholarship, FUELing Excellence Scholarship

TECHNICAL SKILLS

Languages: C++, Python, Java, C, C#, Golang, JavaScript/TypeScript, SQL

Tools/Technologies: React, Node.js, Flask, Django, Spring Boot, Neo4j, Cassandra, PostgreSQL, MySQL, Redis, MongoDB, GCP, AWS, Azure, Docker, Kubernetes, Pandas, PyTorch, Sklearn, Git, Linux

EXPERIENCE

Parasol Lab, University of Illinois

Jan. 2026 - Present

Undergraduate Research Assistant

Urbana, IL

- Contributing to STAPL, an open-source C++ library for HPC on modern parallel systems like clusters and GPUs

Disruption Lab, University of Illinois

Sept. 2025 - Present

Project Manager

Urbana, IL

- Leading 5-member team to deploy distributed LLM inference across GPUs at UIUC MakerLab for a "playground"

Software Engineer

- Developed a VR classroom simulation in Unity/C# for immersive online education with AI teacher interaction
- Implemented real-time multiplayer and proximity chat with Netcode, maintaining <100ms latency for 20+ users
- Monitored with Unity Profiler and optimized network performance via delta compression, quaternion quantization

GEICO

June 2025 - Aug. 2025

Software Development Intern

Chevy Chase, MD

- Created PDLC hub (React, Django, Postgres) for engineers to launch and track projects using stage-gate workflow
- Automated setup via LLM-drafted PRDs and risk scores, with RabbitMQ provisioning Azure DevOps, Slack, repo
- Cut latency by 90% for 3,500+ engineers with Redis caching for queries and endpoints identified via Prometheus
- Deployed to Azure Kubernetes Service using Terraform IaC and Azure Pipelines CI/CD with Nginx load balancing
- Built Python service monitoring 17k+ repos via GraphQL with automated alerting through Slack/email webhooks

Fantasy Trading Room | fftradingroom.com

Sept. 2024 - May 2025

Project Manager & Software Engineer

Urbana, IL

- Led 8-member team using Agile sprints to build scalable fantasy platform with 900k+ views and 500+ users
- Deployed Flask backend on AWS EC2 with CloudWatch monitoring; optimized response times via Redis caching
- Prototyped LLaMA fantasy advice chatbot with QLoRA fine-tuning (HuggingFace / PyTorch) and designed event-driven ML pipeline using Celery for automated retraining triggered by user feedback corrections in Postgres

E42.ai

June 2024 - Aug. 2024

Software Engineering Intern

Remote

- Built LangChain RAG chatbot (E42 LLM, ChromaDB) reducing HR query response time from hours to <2 min
- Developed pandas/scikit-learn evaluation pipeline to score RAG responses against ground truth Q&A pairs
- Created text-to-SQL RAG pipeline using ChromaDB embeddings + schema-level retrieval, enabling natural language database queries; generated 40,000+ synthetic employee records (Python/Faker) matching MySQL

PROJECTS

Distributed Soundboard | Java, C++, Spring Boot, gRPC, PostgreSQL, Apache Kafka, FFmpeg, React, Docker, GCP

- Built async C++ gRPC service with completion queues and CallData states to stream pitch/speed-modified audio chunks (FFmpeg) as processed without disk writes, cutting client memory usage by 80% vs full file downloads
- Handled concurrent audio processing requests using multi-threaded worker pool to prevent resource exhaustion
- Designed Java Spring Boot backend with PostgreSQL and Kafka event streaming to aggregate playback analytics

Data Extraction Program | Scala, Python, Apache Spark, Spark NLP

- Built distributed Spark NLP pipeline using NerDLModel with BERT embeddings to extract names, address, contact info from unstructured documents for Customer Data Integration, achieving 90% F1 score on validation set